

DIEGO ALEJANDRO BOTINA

Software Engineer | System Architect | Full-Stack Developer

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PROFESSIONAL SUMMARY

Software Engineer and System Architect with academic excellence (highest ICFES score in generation, graduated with honors). Specialized in full-stack development using Java, Python, JavaScript, TypeScript, and modern frameworks (Spring Boot, React, .NET, Avalonia UI). Proven expertise in building scalable desktop applications, real-time systems, and cloud-native solutions (AWS, Supabase, Cloudflare). Strong foundation in software architecture patterns (MVVM, Microservices, REST APIs), Agile methodologies (Scrum), and DevOps practices (CI/CD, Docker). Seeking a Software Engineer or Full-Stack Developer role to deliver high-impact, maintainable solutions with clean code and architectural rigor.

CORE COMPETENCIES

- Languages: Java, Python, JavaScript, TypeScript, C#, C++, SQL, HTML5, CSS3
- Frameworks: Spring Boot, React, .NET 9.0, JavaFX, Avalonia UI, Qt6, Astro, Deno/Fresh
- Cloud & DevOps: AWS (S3, EBS), Docker, GitHub Actions (CI/CD), Cloudflare Pages, Nginx, Supabase
- Databases: PostgreSQL, SQLite, Firebase, Supabase (PostgreSQL), relational & NoSQL design
- Architecture: MVVM, MVC, Microservices, REST APIs, Clean Architecture, SOLID principles, Design Patterns
- Practices: Agile/Scrum, TDD, Clean Code, Code Review, Documentation, Performance Optimization
- Tools: Git, Maven, JUnit, Postman, Figma, Linux/Windows, Cross-platform development
- Domains: Real-time systems, Desktop applications, Web development, Audio DSP, Security (CSP, HSTS, Tor)

PROFESSIONAL EXPERIENCE

Full-Stack Developer & Game Architect | Arcade Maze Chomper Project | Jan 2026 – Present

Cross-platform desktop game development (.NET 9.0, Avalonia UI) for Linux/Windows

- *Engineered real-time multiplayer system supporting 5 concurrent players with global leaderboard using WebSocket-like synchronization*
- *Designed and implemented ghost AI behaviors (Blinky, Pinky, Inky, Clyde) with pathfinding algorithms, improving gameplay engagement*
- Applied MVVM architecture and SQLite for persistent high-score tracking, ensuring clean separation of concerns

Tech: C#, .NET 9.0, Avalonia UI, SQLite, MVVM, Git

Security-Focused Web Developer | OnionForge Open-Source Initiative | Nov 2025 – Present

Open-source platform providing secure .onion static site templates with privacy-by-design

- *Implemented Docker containerization with Nginx reverse proxy and automatic .onion address generation, reducing deployment time by 70%*
- *Enforced security headers (CSP, HSTS, X-Frame-Options) and end-to-end encryption, achieving A+ security rating*
- Created reusable template system adopted by privacy-focused developers in Tor community

Tech: HTML5, CSS3, JavaScript, Docker, Nginx, Tor, Security Best Practices

Backend & Desktop Developer | TaskForge Collaborative Platform | Aug 2025 – Dec 2025

Real-time collaborative task management system with role-based access control (RBAC)

- *Engineered conflict-free synchronization system for concurrent updates, maintaining 100% data consistency across clients*

- Implemented MVVM architecture with JavaFX for responsive UI and SQLite for offline-first capability

Tech: Java, JavaFX, SQLite, MVVM, Concurrency Patterns, Git

Python Developer & DSP Engineer | AudioEnhancer Toolkit | May 2025 – Aug 2025

Production-level audio processing library with professional-grade DSP algorithms

- *Implemented software design patterns (Facade, Strategy, Singleton, Factory) to create extensible, testable architecture*
- *Developed noise reduction (Spectral Gate), normalization (Peak/RMS/LUFS), and format conversion with 40% faster processing vs baseline*

Tech: Python, Digital Signal Processing (DSP), OOP, Design Patterns, NumPy

Frontend Developer & DevOps Engineer | CodeWithBotina Technical Blog | Feb 2025 – Present

High-performance static blog with edge computing and serverless backend

- *Achieved 98/100 Lighthouse performance score and <500ms TTFB using Astro 5, Deno/Fresh, and Cloudflare Pages edge deployment*
- *Integrated Supabase (PostgreSQL) for dynamic content with automatic ISR, reducing server costs by 90%*

Tech: Astro 5, Deno/Fresh, Supabase, PostgreSQL, Cloudflare Pages, CI/CD

EDUCATION

Bachelor's in Commercial Software Engineering – Concentration: Design and Architecture

Jala University | Buga, Colombia | Jan 2025 – Present (Expected Graduation: 2028)

Relevant Coursework: Systems Design, Software Architecture, Advanced Programming, Algorithm Design, Database Systems, Cloud Computing

Computer Systems Technician – Graduated with Honors

Discenter | Buga, Colombia | Sep 2022 – May 2024

High School Diploma – Highest ICFES Score (National Exam) & Academic Honors

Institución Educativa Nuestra Señora de Fátima | Buga, Colombia | 2019 – 2024

Dual Technical Certification: Agricultural Technician & Local Tourism Operations Technician (SENA)

CERTIFICATIONS & ACHIEVEMENTS

- AWS Educate: Getting Started with Storage – Amazon Web Services | Dec 2025
- Programming Fundamentals – MinTIC, Colombian Government | Mar 2023 – May 2023
- *Highest ICFES Score in Generation (Colombian National University Entrance Exam)*
- *Graduated with Honors in all academic stages (Technical & High School)*
- 15+ public repositories on GitHub with 50+ combined stars | Active technical content creator

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, C#, C++, SQL, HTML5, CSS3

Frameworks & Libraries: Spring Boot, React, .NET 9.0, JavaFX, Avalonia UI, Qt6, Astro, Deno/Fresh, jQuery

Cloud & DevOps: AWS (S3, EBS, EC2), Docker, GitHub Actions (CI/CD), Cloudflare Pages, Nginx, Supabase, Serverless

Databases: PostgreSQL, SQLite, Firebase, Supabase (PostgreSQL), relational modeling, query optimization

Architecture & Design: MVVM, MVC, Microservices, REST APIs, Clean Architecture, SOLID, Design Patterns (GoF), UML

Methodologies: Agile/Scrum, TDD, Clean Code, Code Review, Git Flow, Documentation, Performance Profiling

Tools & Platforms: Git, Maven, JUnit, Postman, Figma, Linux (Ubuntu), Windows, Cross-platform development

Specialized Knowledge: Digital Signal Processing (DSP), Real-time systems, Security (CSP/HSTS), Tor/.onion, Audio engineering

LANGUAGES

- Spanish – Native (C2)
- English – Professional Working Proficiency (B2) – Fluent technical reading, writing, and documentation